

Supplementary Material

Governing Interruptions After Deterministic Deterioration Scoring

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Supplementary methods

Frozen replay and version boundary

The manuscript reports the Challenge 2019 study frozen at Git commit <https://github.com/sreeram843/curie-prediction-pipeline/commit/1d8a1117151619e2d34822e365d17a00e30e5233>. The setB replay was repeated from that revision with the frozen winner and reproduced the published primary counts. Later scorer revisions are intentionally outside this study. The aggregate v3 holdout sidecar adds denominators, patient-time, stay-level PPV, lead-time quartiles, alert-count distributions, and normalized Challenge utility without replacing v1 or v2 frozen artifacts. Patient-level rows are not stored in the repository.

Partial-SOFA mapping

At each hourly row, the replay retained the most recently observed value for each available variable within that stay; it did not use future rows. No maximum carry-forward age was imposed in the frozen revision. O2Sat was divided by FiO₂, with SaO₂ used when O2Sat was absent. The resulting S/F-like ratio was compared directly with the numeric P/F bands; no validated S/F conversion was applied, so this is a nonstandard respiratory proxy. Because mechanical ventilation was unavailable, respiratory scores of 3 or 4 were not assigned. The cardiovascular component used MAP only. Renal scoring used creatinine only. GCS and urine-output windows were unavailable. At least two scoreable components were required, and threshold-only emission occurred at partial SOFA ≥ 2 .

Table 1: Supplementary Table S1. Component mapping in the frozen Challenge replay. Values at a boundary enter the less severe band unless stated.

Component	Available input	Points
Respiration	O2Sat/FiO ₂ (SaO ₂ fallback)	< 300: 2; 300–< 400: 1; ≥ 400: 0
Coagulation	Platelets (10 ⁹ /L)	< 20: 4; < 50: 3; < 100: 2; < 150: 1; else 0
Liver	Bilirubin (mg/dL)	≥ 12: 4; ≥ 6: 3; ≥ 2: 2; ≥ 1.2: 1; else 0
Cardiovascular	MAP (mmHg)	< 70: 1; otherwise 0
Renal	Creatinine (mg/dL)	≥ 5: 4; ≥ 3.5: 3; ≥ 2: 2; ≥ 1.2: 1; else 0
CNS	GCS	Unavailable; missing

Selection procedure

The original 23-candidate setA sweep used the legacy grace-6 definition: first emission at or before `label_start`+6 hours. It was not the later bounded $-12/+6$ -hour primary-window definition. Passing required governed sensitivity at least naive sensitivity minus 0.10 or at least 0.70, plus an interruptive-to-naive emission ratio no greater than 0.25. Ranking prioritized passing both objectives, lower interruptive ratio, lower emission burden per detected positive stay, and higher governed sensitivity. The winner had 88.4% governed sensitivity under the legacy grace-6 selection estimand. Recalculation under the later primary window yielded 87.6%; these values answer different estimands and are not interchangeable.

Timing sensitivity

The primary setB estimand required any alert in the bounded interval $[\text{label_start} - 12\text{ h}, \text{label_start} + 6\text{ h}]$. The following secondary definitions were computed without changing the frozen governance configuration.

Statistical definitions

Sensitivity was the proportion of label-positive stays with at least one qualifying emission in the selected window. Emissions per detected positive stay divided all emissions in a lane by positive stays detected by that lane; it is not a number of clinically adjudicated pages. Stay-level PPV divided positive stays detected in-window by all stays with any emission. For negative stays, an emission at any recorded hour counted in the denominator. Lead time was `label_start` minus the first governed emission within the primary window. Confidence intervals used 1,000 stay-level percentile-bootstrap replicates with seed 42.

Normalized Challenge utility used emit-hour predictions rather than latching a patient positive after the first emission. It is reported as an additional within-resource metric, not as a hidden-test leaderboard result and not as a clinical utility estimate.

Table 2: Supplementary Table S2. Complete 23-candidate setA selection grid under the legacy grace-6 estimand. Persist and Refrac are minutes; Base and Gate are baseline and page-gate states.

Candidate	Persist	Refrac	Base	Gate	Gov. sens. (%)	Int. sens. (%)	Int. ratio	Selected
grid_p0_r90_b0	0	90	off	on	88.4	37.7	0.123	yes
grid_p0_r120_b0	0	120	off	on	88.4	37.7	0.123	no
dual	0	90	off	on	88.4	37.7	0.123	no
grid_p15_r90_b0	15	90	off	on	86.1	38.6	0.123	no
grid_p15_r120_b0	15	120	off	on	86.1	38.6	0.123	no
grid_p30_r90_b0	30	90	off	on	86.1	38.6	0.123	no
grid_p30_r120_b0	30	120	off	on	86.1	38.6	0.123	no
grid_p0_r60_b0	0	60	off	on	88.4	40.3	0.245	no
grid_p30_r60_b0	30	60	off	on	86.1	40.3	0.245	no
grid_p15_r60_b0	15	60	off	on	86.1	40.3	0.245	no
sensitive	0	120	off	off	88.4	44.1	0.161	no
accuracy	0	90	off	off	88.4	44.1	0.161	no
strict	30	120	on	off	16.9	16.9	0.024	no
grid_p15_r120_b1	15	120	on	on	26.4	26.2	0.042	no
grid_p15_r90_b1	15	90	on	on	26.4	26.2	0.042	no
grid_p30_r90_b1	30	90	on	on	26.4	26.2	0.042	no
grid_p30_r120_b1	30	120	on	on	26.4	26.2	0.042	no
grid_p0_r120_b1	0	120	on	on	29.9	29.7	0.049	no
grid_p0_r90_b1	0	90	on	on	29.9	29.7	0.049	no
balanced	15	60	on	on	26.4	26.2	0.075	no
grid_p15_r60_b1	15	60	on	on	26.4	26.2	0.075	no
grid_p30_r60_b1	30	60	on	on	26.4	26.2	0.075	no
grid_p0_r60_b1	0	60	on	on	29.9	29.7	0.089	no

Table 3: Supplementary Table S3. Secondary setB timing definitions.

Definition	Naive sensitivity (%)	Governed sensitivity (%)
first alert at or before label_start	56.3	56.3
first alert at or before label_start + 6 hours	81.1	81.1
first alert at or before label_start + 12 hours	84.6	84.6
first alert before label_start	54.8	54.8
any alert in [label_start - 12 hours, label_start + 12 hours]	83.3	83.3

Supplementary reproducibility assets

The frozen operating point, resolved study bundle, content hash, timing policy, complete aggregate selection grid, holdout sidecars, comparator results, ablations, miss attribution, bootstrap intervals, and machine-readable tables are included in the repository. The public dataset is available at <https://doi.org/10.13026/v64v-d857>. Rebuild commands and pinned hashes are listed in `paper/REPRODUCIBILITY.md` and the manuscript manifest.